THE WAR ON SNOOPING EYES



HOW A CCTV WORKS

1. Light enters the camera lens, hits a digital sensor and is converted into electrical pulses

2. A microchip called System-on-Chip (SoC) has a hardware processor that works under software instruction. The SoC converts pulses into digital data

➘The software is stored in Flash Memory and Read only Memory (RoM)—known as the Firmware

3. Motion/ face detection apps in the SoC help sift through large amounts of image data

4. Captured data goes to a recorder/ cloud storage

5. The recorder converts data back into video feed for one to watch

HOW IT IS HACKED

SPOOFING & BACKDOOR

In a spoofing attack, hackers connect to a network under the disguise of a trusted device. In a backdoor attack, they search for vulnerabilities, like insecure servers

THE CHINESE THREAT

Most CCTVs are Chinese, and they use SoCs (along with the firmware) made by HiSilicon, which is a part of state-owned tech giant Huawei. If the Chinese firmware has an inbuilt vulnerability, it can easily be hacked

➘When connected to the internet, they can transmit data back to Chinese servers